Mobile Repeater Coverage Along the Susquehanna River / NS Port Road Branch

Terrain-aware VHF/UHF propagation analysis — Mobile Relay to an HT

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Contents

1 Summary

This study evaluates whether a **mobile cross/same-band repeat** operated from a mobile communications vehicle parked at the Muddy Run (Wissler Park) can reach a **handheld (HT) carried on foot** along the Lancaster-side of the Susquehanna River, along Norfolk Southern's Port Road Branch between Holtwood (MP 25.0) and Midway (MP21.7) specifically, plus a further 3-5 mi north and / or south.

The comm vehicle sits essentially on the valley floor — **~102 m below the average surrounding terrain** — which at first glance appears to be poor coverage. It is not for this mission: the HT travels the *same low corridor*, so signal propagates **along the valley** rather than fighting uphill over ridges. All three bands analyzed deliver usable HT coverage across the full hiking range, with one frequency- and direction-dependent exception.

Headline result (terrain + summer foliage):

- **2M (146.1 MHz)** and **AAR VHF (~161.0 MHz)** — solid copy the entire 5 miles, both directions. These are the reliable choices for the repeat leg.
- **GMRS (462.6 MHz)** — solid to ~3 miles both ways; **going north it degrades to marginal/dead past ~5.5 km** in summer foliage. South, GMRS stays usable the whole way.
- **Seasonal effect:** leaf-off (fall/winter) recovers ~4 dB on UHF and ~3 dB on VHF, pulling the GMRS-north dead zone back toward marginal. The wooded corridor measurably improves as the leaves drop.

2 Objective and operational context

The operator hikes along the Susquehanna, monitoring rail (AAR VHF), GMRS, APRS and 2 m repeaters. The comm wagon — with capable roof/fender antennas and good receivers — pulls in distant signals that a handheld cannot. The question is the **outbound relay leg**: when the wagon re-transmits to the HT the operator carries up and down the corridor, *how far does that repeat hold up?*

The comm vehicle is fixed (a riverside park). The requirement is **not** to defeat the valley in all directions — it is to reach the operator along the paths actually hiked, which follow the river/rail corridor and occasionally climb onto the adjacent Lancaster-side high ground.

3 Method

3.1 Propagation engine and terrain

- **Engine:** SPLAT! HD v1.4.2 using the **ITWOM 3.0 / Longley-Rice Irregular Terrain Model**. ITWOM models terrain diffraction and ground reflection; it does **not** model vegetation (handled separately, below).
- **Terrain data: NASADEM 1-arc-second (~30 m)**, void-filled and error-corrected, tile `N39W077` covering the study area. HD (`splat-hd`) tools and 1-arc-second data were used throughout for ridge/diffraction fidelity.
- **Transmitter (com vehicle):** 39.8059656, -76.2963533; antenna 2.0 m AGL (vehicle roof/fender). Ground elevation 40 m AMSL (valley floor; -102 m below average terrain).
- **Receiver (HT):** 1.5 m AGL (head height).
- **ITWOM environment:** earth dielectric 15, conductivity 0.005 S/m, N = 301, radio climate 5 (continental temperate), **vertical polarization** (FM), 90 % of situations / 90 % of time.

3.2 Per-band parameters

Band	Freq (MHz)	ERP (W)	Wagon antenna
GMRS	462.6	45	Laird C150/450C fender collinear
2 m	146.1	50	Antenex B1443 5/8-wave roof
AAR VHF	161.0	55	Laird C150/450C fender collinear

3.3 Corridor tracing

The hiking corridor was traced directly from the DEM by a **low-ground follower**: starting at the wagon, stepping 250 m at a time along the general up-river (NW, ~325°) and down-river (SE, ~145°) bearings, at each step choosing the lowest-elevation neighbor within ±70° so the track hugs the river channel. Waypoints were recorded every ~1 km out to 8 km in each direction (N1–N8, S1–S8). This approximates the rail/river corridor closely without requiring surveyed trail coordinates.

3.4 Point-to-point analysis

For each of the 3 bands x 2 directions x 8 waypoints (**48 links**), a SPLAT! HD path analysis was run from the wagon to the waypoint, extracting ITWOM path loss, terrain-shielding loss, received signal power (dBm), and the dominant propagation mode.

One waypoint (N4, ~4 km north) triggered an ITWOM numerical edge case that hangs before writing its report; its received level is interpolated from the bracketing N3/N5 results and flagged as such in the data.

3.5 Foliage correction (two seasons)

Vegetation loss — absent from ITWOM — was applied as a separate layer using the **Weissberger modified exponential-decay model**, evaluated at two effective foliage depths representing the wooded corridor:

- **Summer (full leaf):** 30 m effective vegetation depth.
- **Leaf-off (fall/winter):** 10 m effective depth (trunks/branches only).

These depths are first-order estimates for an intermittently wooded river/rail corridor and are the dominant assumption in the seasonal comparison. Resulting excess loss:

Band	Summer foliage loss	Leaf-off loss	Seasonal gain
GMRS 462.6	7.9 dB	3.6 dB	**4.3 dB**
2 m 146.1	5.7 dB	2.6 dB	**3.1 dB**
AAR 161.0	5.8 dB	2.7 dB	**3.2 dB**

Higher frequency suffers more foliage loss and therefore gains the most from leaf-off — which is exactly where the GMRS-north edge case lives.

3.6 Usable-signal thresholds (received power at the HT)

Verdict	dBm	Meaning
SOLID	at or above -100	full quieting / easy copy
USABLE	-100 to -110	readable, some noise
MARGINAL	-110 to -118	breaking up
DEAD	< -118	below usable HT sensitivity

4 Results

4.1 Signal vs. distance

The clearest view is received HT signal against corridor distance, per band, under each foliage state, with the usable-signal bands shaded.

Key reads:

- **VHF (2 m, AAR) stays in the SOLID band the full 8 km both directions**, in every foliage state. These bands own the corridor.
- **GMRS holds SOLID to ~3 miles**, then diverges by direction:
  - **South:** remains USABLE to the full 8 km (gentler, single-horizon terrain).
  - **North:** the up-river double-horizon terrain drives increasing shielding (up to ~63 dB); GMRS crosses into MARGINAL near 5.5 km and, **with summer foliage, into DEAD beyond ~6.5 km**.
- **The three-panel progression** (terrain -> summer -> leaf-off) shows the whole family of curves shifting down ~6–8 dB in summer and recovering in leaf-off — visually confirming the seasonal improvement.

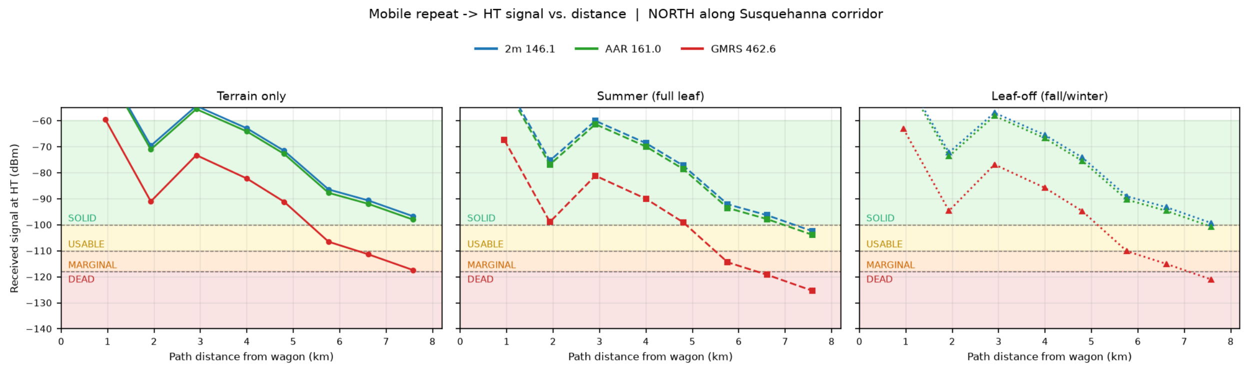

Figure 1: Signal vs. distance — NORTH (up-river). Left: terrain only; center: summer full-leaf; right: leaf-off.

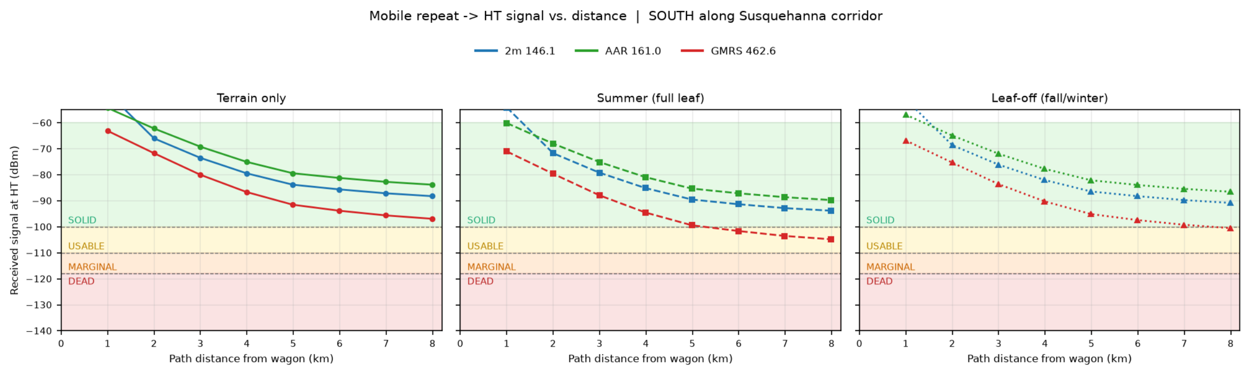

Figure 2: Signal vs. distance — SOUTH (down-river). Left: terrain only; center: summer full-leaf; right: leaf-off.

4.2 GMRS north — the one problem area, and its seasonal recovery

North waypoint	Path km	dBm terrain	dBm summer	dBm leaf-off	Summer verdict	Leaf-off verdict
N5	4.80	-91.1	-99.0	-94.7	SOLID	SOLID
N6	5.76	-106.5	-114.4	-110.1	MARGINAL	MARGINAL
N7	6.61	-111.3	-119.2	-115.0	**DEAD**	MARGINAL
N8	7.58	-117.4	-125.3	-121.0	**DEAD**	DEAD

Leaf-off recovers N7 from DEAD back to MARGINAL — a real, usable operational difference for fall/winter train chasing.

4.3 Coverage maps

SPLAT! area-coverage maps (ITWOM signal-power contours over shaded terrain relief, HT at 1.5 m AGL, 5 km radius) confirm the corridor-following behavior — coverage extends along the low ground and is shadowed by the surrounding rises. The wagon (white star) and corridor waypoints (colored by summer HT verdict) are overlaid; the signal-level color key is at right.

The same corridor is shown below on a NASADEM elevation-contour base, making the valley structure explicit — the wagon sits in the drainage and the waypoints follow the low ground north and south. Waypoint markers are colored by the summer HT verdict for that band.

5 Discussion

- **The valley "hole" is not the enemy for a corridor mission.** The -102 m below-average-terrain figure describes an omnidirectional disadvantage, but the operator and the wagon share the drainage. Signal follows the corridor; the metric that mattered was *along-corridor* diffraction, not average-terrain clearance.
- **Frequency choice is the lever.** All paths are diffraction-dominated (no true line-of-sight). Lower frequency diffracts better, so **VHF outperforms UHF** along the wooded, terrain-broken corridor. For maximum reliable range — especially north — **operate the repeat leg on 2 m or AAR VHF**. Reserve GMRS for the ~3-mile solid zone and the southern corridor.
- **Season matters, predictably.** Summer foliage costs ~6–8 dB; leaf-off returns most of it. The practical upshot: **GMRS reach north improves in fall/winter**, the prime train-chasing season anyway.

6 Limitations

- **Foliage depths are estimates**, not measured. The 30 m / 10 m summer/leaf-off effective depths are first-order; actual loss varies with specific tree density and path geometry. The *relative* seasonal comparison is more robust than the absolute dB.
- **Weissberger is an empirical bulk model**, applied uniformly along each path; it does not resolve where along the route the trees actually are.
- **Antenna patterns are not yet modeled** — ERP is treated as into an idealized pattern. The fender-mounted C150/450C in particular is slightly directional in the real install (vehicle-body interaction), not captured here.
- **N4 (north, ~4 km) is interpolated** due to an ITWOM edge case.
- **ITWOM 90/90 confidence** — results are the 90 %-of-situations / 90 %-of-time prediction, appropriately conservative.

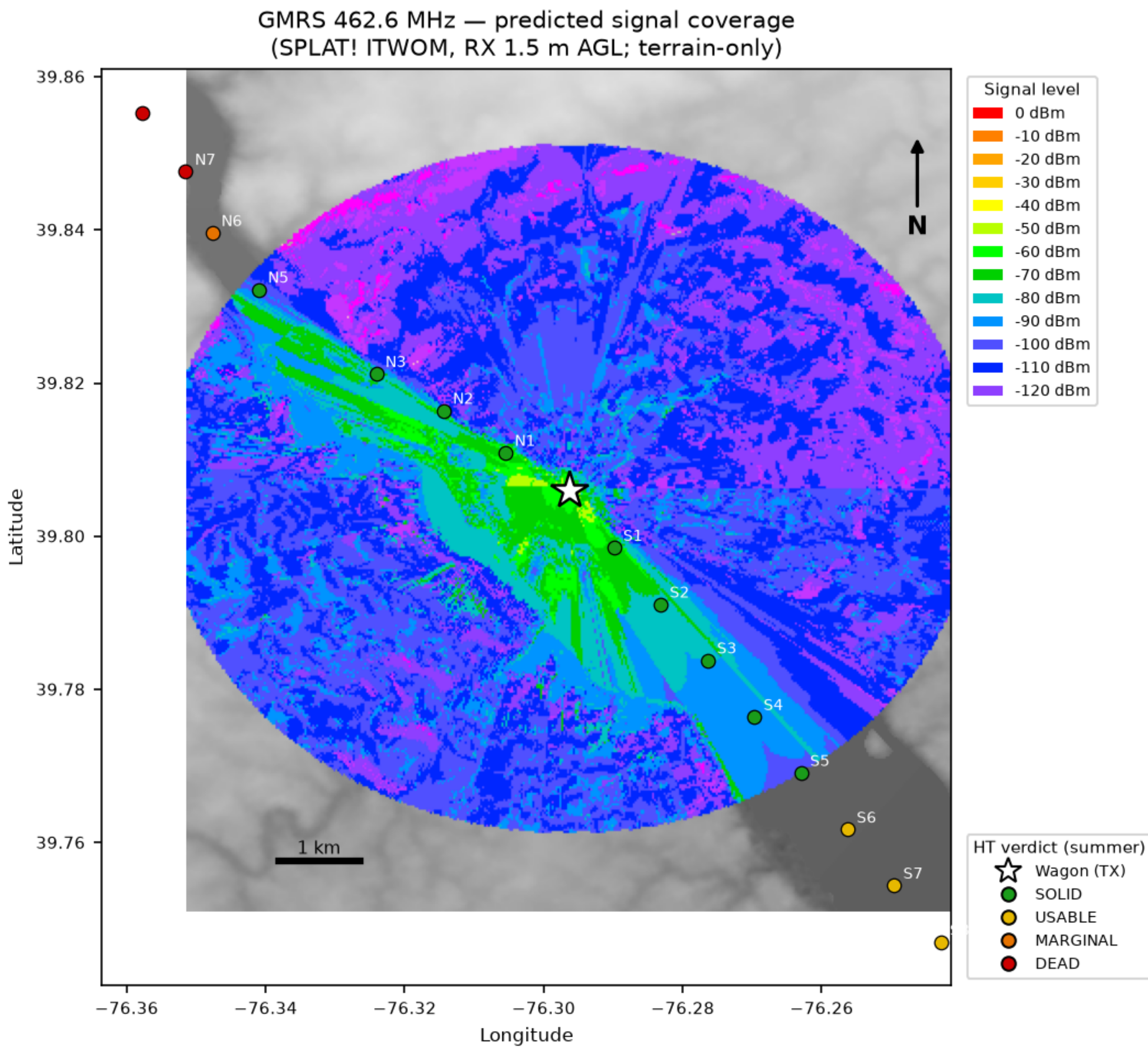

Figure 3: GMRS 462.6 MHz predicted signal coverage over terrain relief.

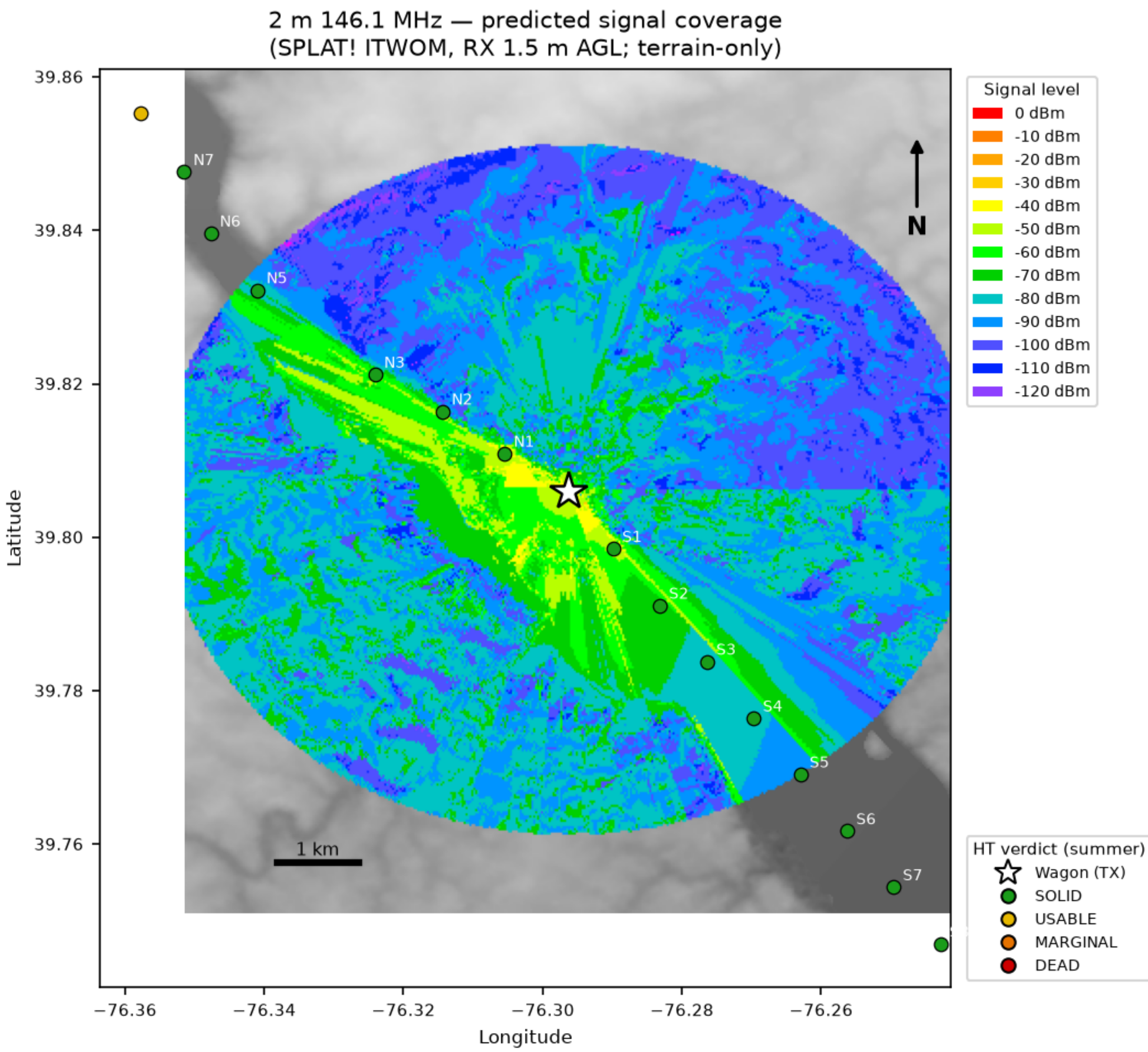

Figure 4: 2 m 146.1 MHz predicted signal coverage over terrain relief.

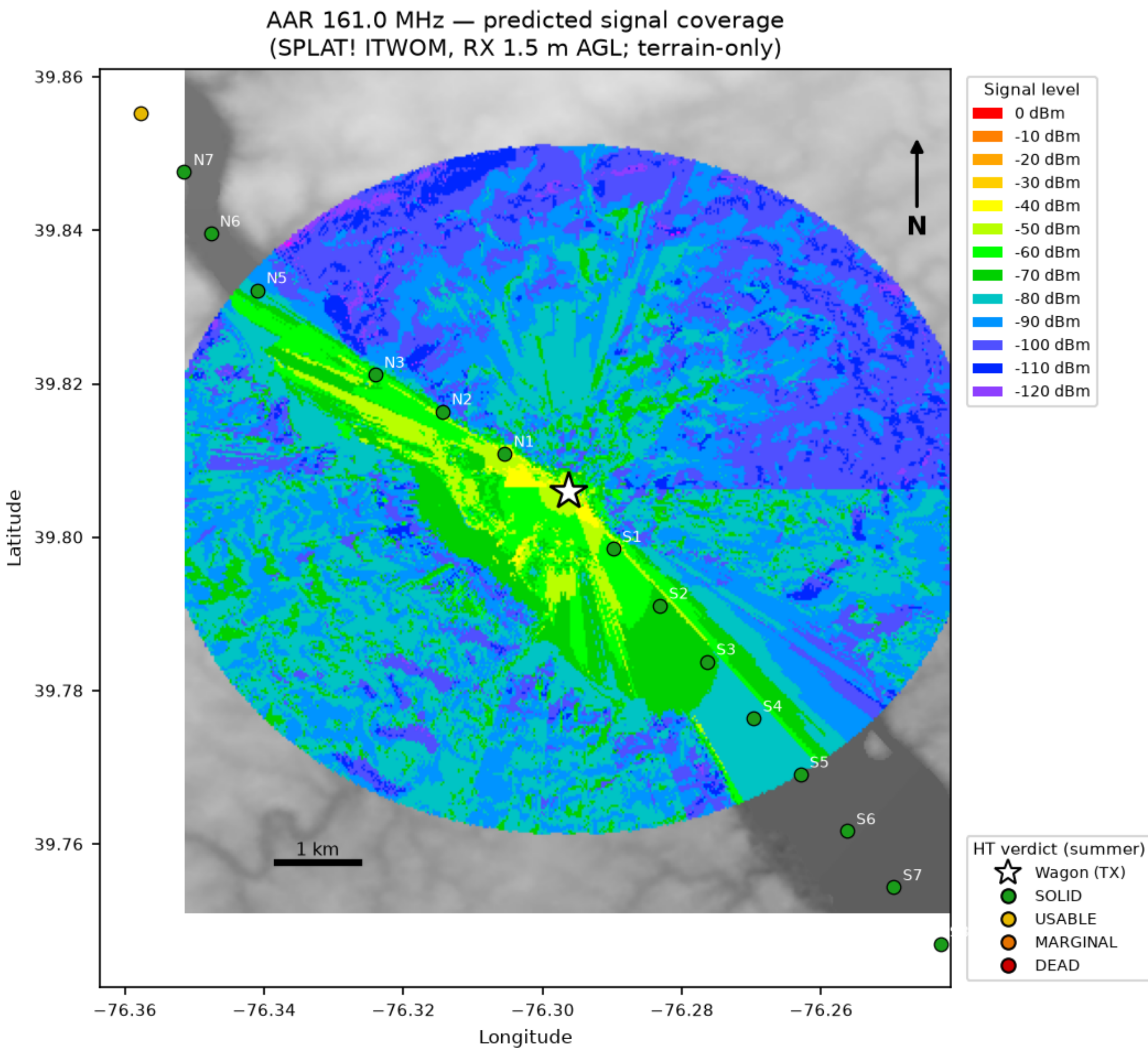

Figure 5: AAR 161.0 MHz predicted signal coverage over terrain relief.

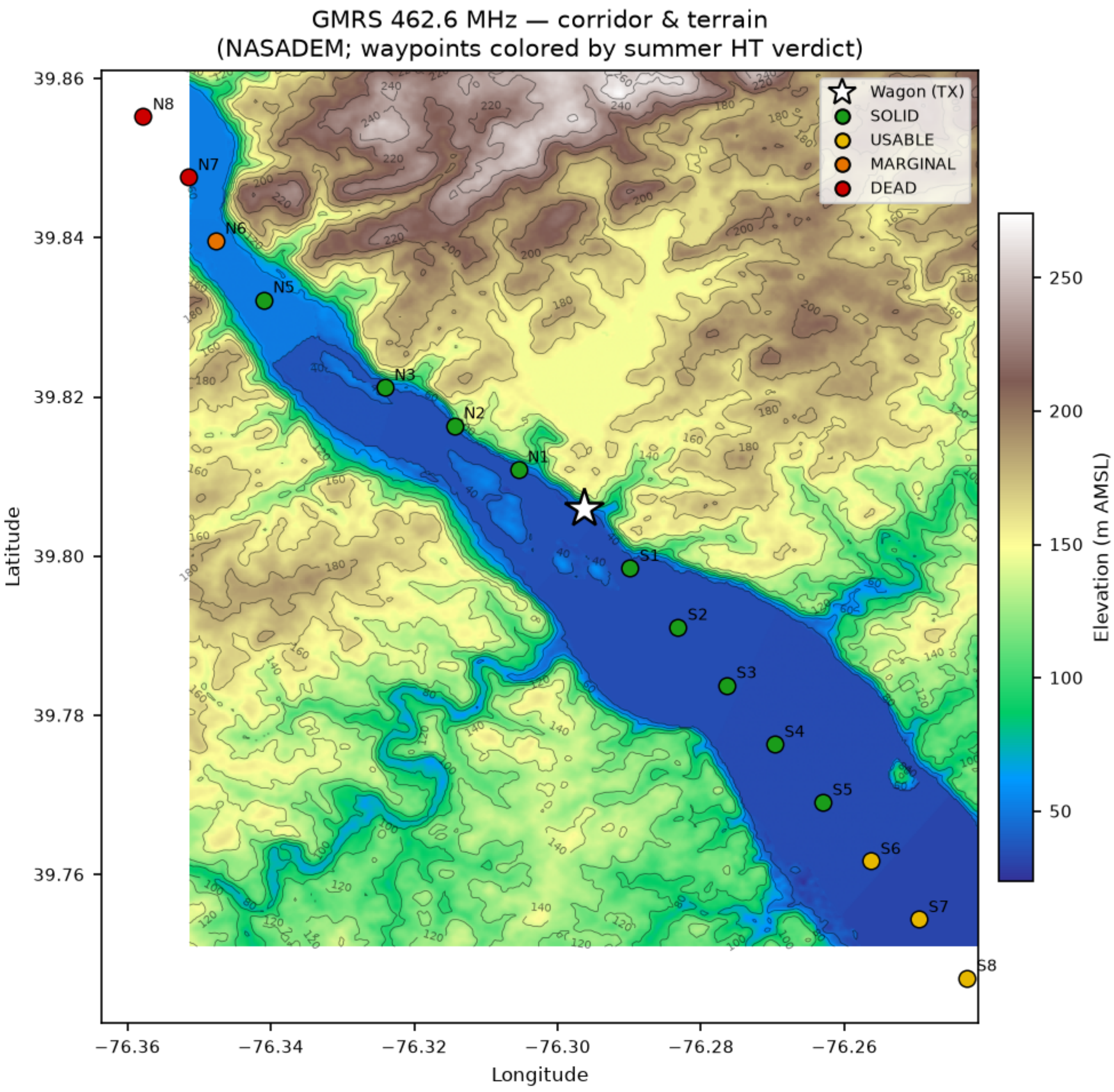

Figure 6: GMRS corridor & terrain; waypoints by summer verdict.

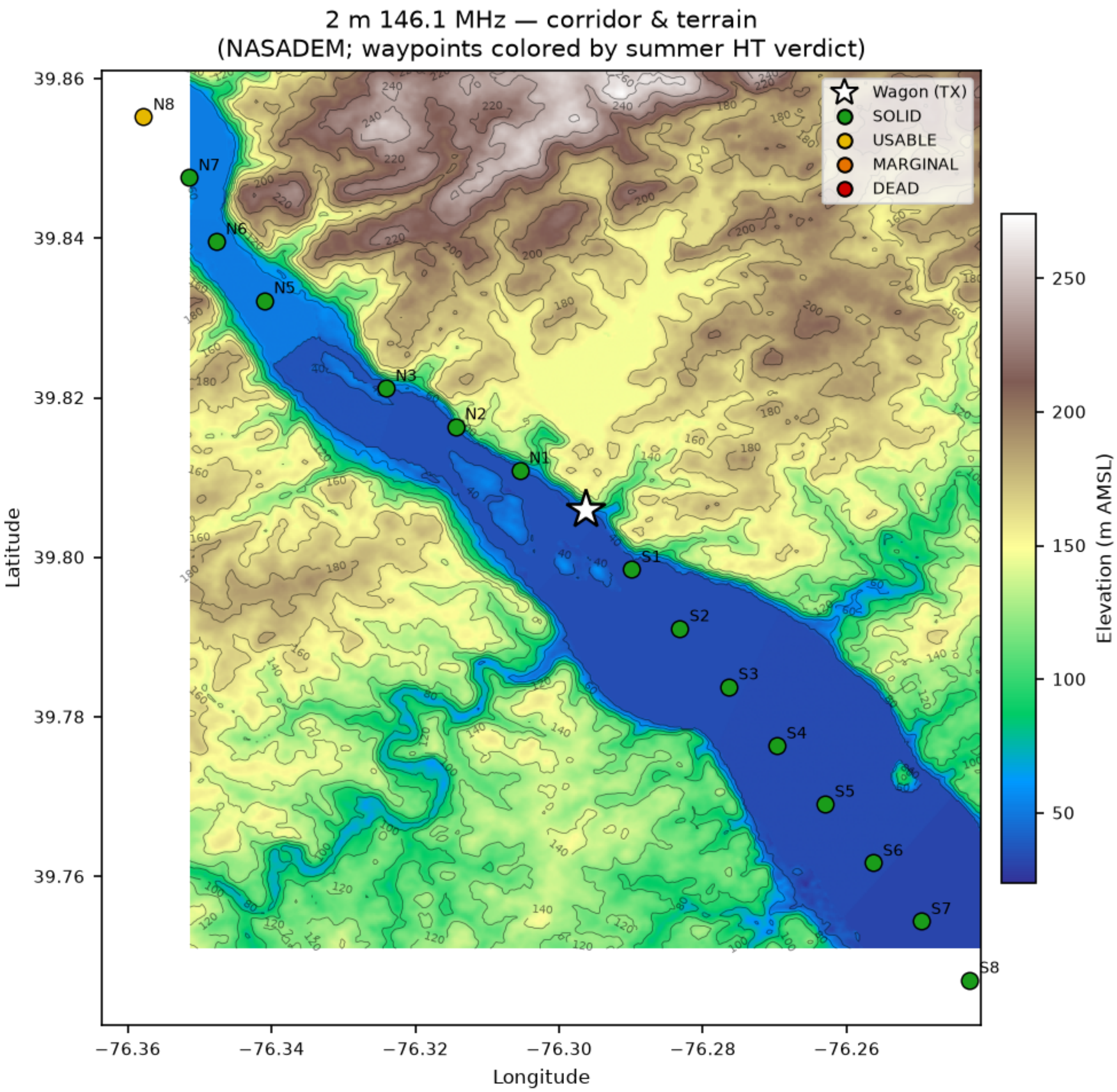

Figure 7: 2 m corridor & terrain; waypoints by summer verdict.

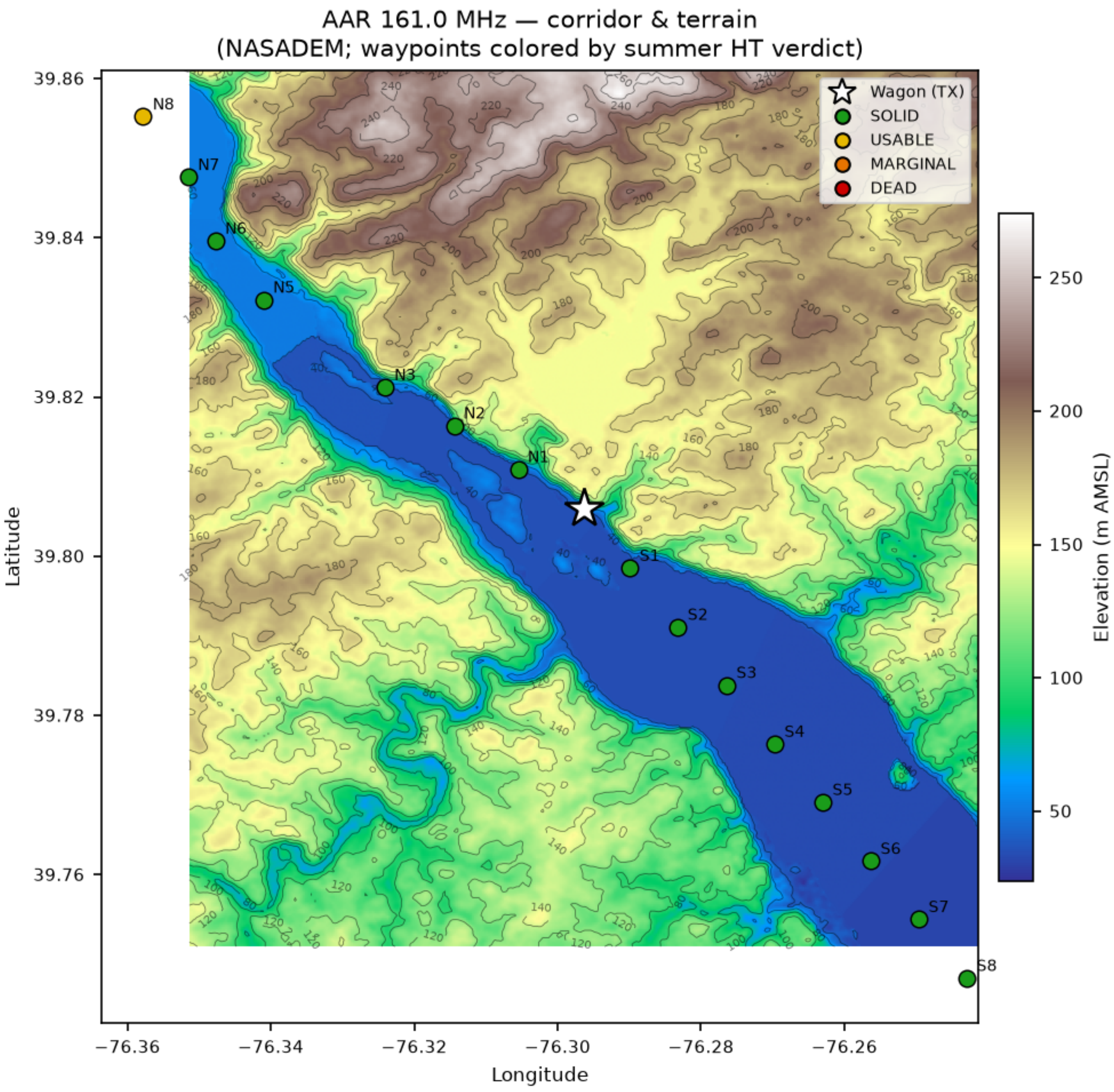

Figure 8: AAR corridor & terrain; waypoints by summer verdict.

7 Reproducibility

All inputs, code, and derived data are in the `Splat_RF_Analysis` repository:

- Study definition: `studies/muddy_run_corridor/study.yml`
- Corridor trace: `splat_rf/corridor/trace.py`
- Results build (parse + foliage): `splat_rf/corridor/build_results.py` -> `studies/muddy_run_corridor/results.c`
- Plots: `splat_rf/corridor/plot_signal.py`
- KML export: `splat_rf/corridor/build_kml.py` -> `reports/figures/kml/`
- PostGIS load: `splat_rf/corridor/load_postgis.py` (schema `rf_analysis` in `gis_dev`; 16 sites + 48 path_profiles, QGIS-loadable)
- Terrain (not in repo, per big-data-on-/data rule): `/data/splat/sdf/`

Render this report to PDF:

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pandoc reports/muddy_run_corridor.md -o reports/muddy_run_corridor.pdf \
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